Chapter: 1

Measures of Central Tendency and Measures of Dispersion

❖ Central Tendency:

The tendency of clustering other points to a value that is located at central is called central tendency. Here we represent a set of data by a single value.

❖ Different Measures of Central Tendency:

The following are the important measures of central tendency which are generally used in business:

1. Arithmetic mean
2. Geometric mean
3. Harmonic Mean
4. Median
5. Mode

Arithmetic mean:

Calculation of Arithmetic Mean- Ungrouped Data:

The **arithmetic mean**, often simply referred to as **mean**, is the total of the values of a set of observations divided by their total number of observations. Thus, if $X_{1,}X_{2,}$………………X_N represents the values of N items or observations, the arithmetic mean denoted by $\overline{X}$ is defined as:

$$\overline{X} = \frac{X_1 + X_2 + \dots\dots\dots + X_N}{N} = \frac{\sum_{i=1}^{N} X_i}{N}$$

Example 1:

The monthly income (in Tk's) of 10 persons working in a firm is as follows:

14870	14930	15020	14460	14750	14920	15720
15160	14680	14890				

Find **average** monthly income.

Example 2:

Sun Com is studying the number of minutes used monthly by clients in a particular cell phone rate plan. A random sample of 12 clients showed the following number of minutes used last month.

90	77	94	89	119	112
91	110	92	100	113	83

What is the **arithmetic mean** number of minutes used?

Calculation of Arithmetic Mean-Grouped Data:

When the direct method is used

$$\overline{X} = \frac{\sum fX}{N}$$

Where,

X=mid-point of various classes

f= the frequency of each class

N=the total frequency

Note: For computing mean in the case of grouped data the midpoints of the various classes are taken as representative of that particular class. The reason is that when the data are grouped, the exact frequency with which each of the variables occurs in the distribution is unknown.

Example 3:

The following are the figures of profits earned by 1400 companies during 1999-2000.

Profits (Tk.lakhs)	No. of companies	Profits (Tk.lakhs)	No. of companies
200-400	500	1000-1200	100
400-600	300	1200-1400	80
600-800	280	1400-1600	20
800-1000	120		

Calculate the **average profits** for all companies.

Formula:

If we have the arithmetic mean and the number of observations in two or more two related groups, we can compute the combined average of these groups by applying the following formula.

$$\overline{X_{12}} = \frac{N_1 \overline{X_1} + N_2 \overline{X_2}}{N_1 + N_2}$$

Where,

$\overline{X_{12}}$ =Combined mean of the two groups

$\overline{X_1}$ =Arithmetic mean of the first group
$\overline{X_2}$ =Arithmetic mean of the second group
N_1 =Number of observations in the first group
N_2 = Number of observations in the second group

*** If we have to find out the combined mean of three series, the above formula can be extended as follows:

$$\overline{X_{123}} = \frac{N_1\overline{X_1} + N_2\overline{X_2} + N_3\overline{X_3}}{N_1 + N_2 + N_3}$$

Example 4:
There are two branches of a company employing 100 and 80 persons respectively. If the arithmetic mean of the monthly salaries paid by the two branches is tk.1570 and tk.1750 respectively, find the arithmetic mean (average) of the salaries of the employees of the company as a whole.

❖ Weighted Arithmetic Mean:
One of the limitations of the arithmetic mean is that it gives equal importance to all the observations. But there are cases where the relative importance of all the different observations is not the same. When that is so, we compute the weighted arithmetic mean.

The term 'weight' stands for the relative importance of the different observations. The formula for computing weighted arithmetic mean is

$$\overline{X}_w = \frac{\sum WX}{\sum W}$$

Where,
$\overline{X}_w$ =Represents the weighted arithmetic mean
X= The variable
W = Weights attached to the variable X.

The weighted mean is especially useful in the problems relating to the construction of index numbers and standardized birth and death rates.

Example 5:
The Cater Construction company pays its hourly employees $16.50,$17.50, or $18.50 per hour. There are 26 hourly employees, 14 are paid at the $16.50 rate, 10

at the \$17.50 rate, and 2 at the \$18.50 rate. What is the mean hourly paid for the 26 employees?

Example 6:

Suppose that, the nearby Wendy's Restaurant sold medium, large, and Biggie-sized soft drinks for \$0.90, \$1.25, and \$1.50 respectively of the last 10 drinks sold, 3 were medium, 4 were large and 3 were Biggie-sized. Find the mean price of the last 10 drinks sold.

Solved Examples

Q. 1: The marks obtained by 6 students in a class test are 20, 22, 24, 26, 28, 30. Find the mean.

Solution:

Therefore, mean = 25

Q. 2: If the arithmetic mean of 14 observations 26, 12, 14, 15, x, 17, 9, 11, 18, 16, 28, 20, 22, 8 is 17. Find the missing observation.

Solution:

Given 14 observations are: 26, 12, 14, 15, x, 17, 9, 11, 18, 16, 28, 20, 22, 8

Arithmetic mean = 17

We know that,

Arithmetic mean = Sum of observations/Total number of observations

17 = (216 + x)/14

17 x 14 = 216 + x

216 + x = 238

x = 238 – 216

x = 22

Therefore, the missing observation is 22.

Method 1: Direct Method for Calculating Mean

To find mean using direct method, we can use the following steps:

Step 1: For each class, find the class mark x_i, as

$$x=1/2(\text{lower limit} + \text{upper limit})$$

Step 2: Calculate $f_i.x_i$ for each i.

Step 3: Use the formula **Mean = $\sum(f_i.x_i)/\sum f_i$.**

Example I (discrete grouped data): Find the arithmetic mean of the following distribution:

x	10	30	50	70	89
f	7	8	10	15	10

Solution:

x_i	f_i	x_if_i
10	7	10×7 = 70
30	8	30×8 = 240
50	10	50×10 = 500
70	15	70×15 = 1050
89	10	89×10 = 890
Total	$\sum f_i$=50	$\sum x_if_i$=2750

Add up all the (x_if_i) values to obtain $\sum x_if_i$. Add up all the f_i values to get $\sum f_i$

Now, use the arithmetic mean formula.

$\bar{x} = \sum x_if_i / \sum f_i = 2750/50 = 55$

Arithmetic mean = 55. The above problem is an example of discrete grouped data.

Method 2: Assumed – Mean Method for Calculating Mean

For calculating the mean in such cases, we proceed as under.

Step 1: *For each class interval, calculate the class mark x by using the formula:* ***x_i = 1/2 (lower limit + upper limit).***

Step 2: *Choose a suitable value of mean and denote it by A. x in the middle as the assumed mean and denote it by A.*

Step 3: *Calculate the deviations d_i = (x, -A) for each i.*

Step 4: *Calculate the product (f_i x d_i) for each i.*
Step 5: *Find n = $\sum f_i$*

Step 6: *Calculate the mean, x, by using the formula:* ***X = A + $\sum f_i d_i / n$.***

Example II (continuous class intervals): Let's try finding the mean of the following distribution:

Class - Interval	15-25	25-35	35-45	45-55	55-65	65-75	75-85
Frequency	6	11	7	4	4	2	1

When the data is presented in class intervals, the mid-point of each class (also called class mark) is considered for calculating the arithmetic mean.

The formula for mean remains the same as discussed above.

Class-Interval	Class Mark (x_i)	Frequency (f_i)	x_if_i
15-25	20	6	120
25-35	30	11	330
35-45	40	7	280
45-55	50	4	200
55-65	60	4	240
65-75	70	2	140
75-85	80	1	80
	Total	35	1390

$\bar{x} = \sum x_if_i / \sum f_i = 1390/35 = 39.71$. We have, $\sum f_i = 35$ and $\sum x_if_i = 35$

Arithmetic mean = 39.71

Assumed Mean Method

The short-cut method is called as **assumed mean method** or **change of origin method**. The following steps describe this method.

Step1: Calculate the class marks (mid-point) of each class (x_i).

Step2: Let **A** denote the assumed mean of the data.

Step3: Find deviation (di) = $x_i – A$

Step4: Use the formula:

$\bar{x} = A + (\sum f_id_i/\sum f_i)$

Example: Let's understand this with the help of the following example. Calculate the mean of the following using the short-cut method.

Class - Intervals	45-50	50-55	55-60	60-65	65-70	70-75	75-80
Frequency	5	8	30	25	14	12	6

Solution: Let us make the calculation table. Let the assumed mean be **A** = 62.5

Note: A is chosen from the x_i values. Usually, the value which is around the middle is taken.

Class-Interval	Classmark / Mid-points (x_i)	f_i	$d_i = (x_i - A)$	$f_i d_i$
45-50	47.5	5	47.5-62.5 =-15	-75
50-55	52.5	8	52.5-62.5 =-10	-80
55-60	57.5	30	57.5-62.5 =-5	-150
60-65	62.5	25	62.5-62.5 =0	0
65-70	67.5	14	67.5-62.5 =5	70
70-75	72.5	12	72.5-62.5 =10	120
75-80	77.5	6	77.5-62.5 =15	90
		$\sum f_i=100$		$\sum f_i d_i= -25$

Now we use the formula,

$\bar{x} = A + (\sum f_i d_i / \sum f_i) = 62.5 + (-25/100) = 62.5 - 0.25 = 62.25$

∴ Arithmetic mean = 62.25

Step Deviation Method

This is also called the **change of origin or scale method.** The following steps describe this method:

Step 1: Calculate the class marks of each class (x_i).

Step 2: Let **A** denote the assumed mean of the data.

Step 3: Find $u_i = (x_i - A)/h$, where h is the class size.

Step 4: Use the formula to find arithmetic mean:

$\bar{x} = A + h \times (\sum f_i u_i / \sum f_i)$

Method 3: Step-Deviation Method for Calculating Mean

When the values of x, and f are large, the calculation of the mean by the above methods becomes tedious. In such cases, we use the step-deviation method, given below.

Step 1: *For each class interval, calculate the class mark x, where* ***X = 1/2 (lower limit + upper limit).***

Step 2: *Choose a suitable value of x, in the middle of the x, column as the assumed mean and denote it by A.*

Step 3: *Calculate h = [(upper limit) – (lower limit)], which is the same for all the classes.*

Step 4: *Calculate $u_i = (x_i - A)/h$ for each class.*

Step 5: *Calculate fu for each class and hence find* ***$\sum(f_i \times u_i)$.***

Step 6: *Calculate the mean by using the formula:* ***$x = A + \{h \times \sum(f_i \times u_i) / \sum f_i\}$***

Example: Consider the following example to understand this method. Find the arithmetic mean of the following using the step-deviation method.

Class Intervals	0-10	10-20	20-30	30-40	40-50	50-60	60-70	Total
Frequency	4	4	7	10	12	8	5	50

Solution: To find the mean, we first have to find the class marks and decide A (assumed mean). Let A = 35 Here h (class width) = 10.

C.I.	x_i	f_i	$u_i = x_i - Ah$ $x_i - Ah$	$f_i u_i$
0-10	5	4	-3	4 x (-3)=-12
10-20	15	4	-2	4 x (-2)=-8
20-30	25	7	-1	7 x (-1)=-7
30-40	35	10	0	10 x 0= 0
40-50	45	12	1	12 x 1=12
50-60	55	8	2	8 x 2=16
60-70	65	5	3	5 x 3=15
Total		∑fi=50		∑fiui=16

Using arithmetic mean formula:

$\bar{x} = A + h \times (\sum f_i u_i / \sum f_i) = 35 + (16/50) \times 10 = 35 + 3.2 = 38.2$

Arithmetic mean = 38.

Example: Find the mean of the following frequency distribution:

Class	50-70	70-90	90-110	110-130	130-150	150-170
Frequency	18	12	13	27	8	22

2. Median:

The median is the middlemost observation when the observations are arranged in ascending or descending order of magnitude. That is, half of the observations in a set of data are lower than it and half of the observations are greater than it.

As distinct from the arithmetic mean which is calculated from the value of every observation in the series, the median is what is called a positional average. The term 'position' refers to the place of a value in a series. The place of the median in a series is such that an equal number of observations lie on either side of it.

For **example**, if the income of five persons is Tk.1000, 1200, 1500, 1600, 1800 then the median income would be Tk. 1500. Changing any or both of the first two values with any other numbers with a value of 1500 or less and/or changing any of the last two values to any other values with values of 1500 and more, would not affect the value of the median which would remain 1500.

In contrast, in the case of arithmetic mean, the change in the value of a single observation would cause the value of the mean to be changed.

Calculation of Median –Ungrouped Data:

For ungrouped data, if n is odd then the median is $\frac{n+1}{2}$th observation, if n is even then the median is the AM of $\frac{n}{2}$ th and $(\frac{n}{2}+1)$th observation.

Example 7:

From the following data of wages of 7 workers, compute the median wage:

Wages (in Rs.): **1600 1650 1580 1690 1660 1606 1640**

Example 8:

The flowing table gives the monthly income of 12 families in a village.

H.No	Monthly income(TK)	H.No	Monthly income(TK)
1	587	7	805
2	693	8	907
3	595	9	763
4	780	10	865

5	840	11	768
6	760	12	894

Calculate the **median income**.

Calculation of Median–Grouped Data:

For grouped data, the median is given by,

$$\mathbf{Me = L +} \frac{\left(\frac{n}{2} - F_c\right)}{f_m} \times c$$

Where, n= total frequency

F_c= cumulative frequency of pre-median class

f_m= frequency of the median class

c= class width

L= lower class boundary of median class.

When the class intervals are arranged in descending order, for computation of median by this formula, the only change in the identification of F values, all other quantities remaining same. F now stands for the sum of all frequencies below the upper limit of the median class.

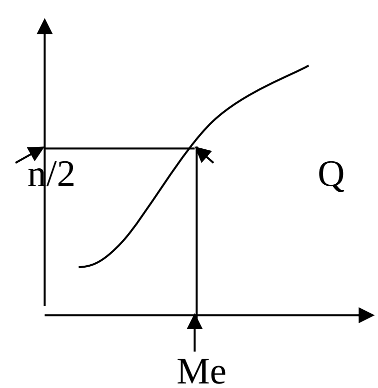
Example 9:1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (yrs)	**No. of workers**	**Age (yrs)**	**No. of workers**
18-22	120	38-42	184
22-26	125	42-46	162
26-30	280	46-50	86
30-34	260	50-54	75
34-38	155	54-58	53

Calculate the **median age**.

Example 10:
Calculate the median from the following data:

Value	**Frequency**
0-10	4
10-20	12
20-30	24
30-40	36
40-50	20
50-60	16
60-70	8
70-80	5



Mode:

The mode is the value of the variable that occurs most frequently, i.e. for which the frequency is maximum. It may be used for both categorical and numerical data. Graphically, it is the value on the X-axis below the peak, or highest point, of the frequency curve as we can see from the flowing diagram. When **speed** is the most important consideration, the mode would be the preferred measure of central tendency.

Calculation of mode –Ungrouped data:

For determining mode count the numbers of items the various values repeat themselves and the value which occurs the maximum number of times is the modal value.

Example 11:

The following figures relate to the preferences with regard to size of screen in inches of T.V sets of 30 persons selected at random from a locality. Find the modal size of the T.V screen.

12	20	12	24	27	20	12	20	27	24
24	20	12	20	24	27	24	24	20	24
24	20	24	24	12	24	20	27	24	24

Calculation of mode- Grouped data:

For grouped data from a frequency distribution the mode can be obtained as:

$$\boldsymbol{M_o = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c}$$

Where,

L= Lower class boundary of the modal class,

f_1 = Frequency of the modal class,

f_0 = Frequency of the pre-modal class,

f_2 = Frequency of the post-modal class.

An **alternative formula** for calculating mode is

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i$$

Where,

L=Lower limit of the modal class

Δ_1 = The difference between the frequency of the modal class and the frequency of the pre-modal class.

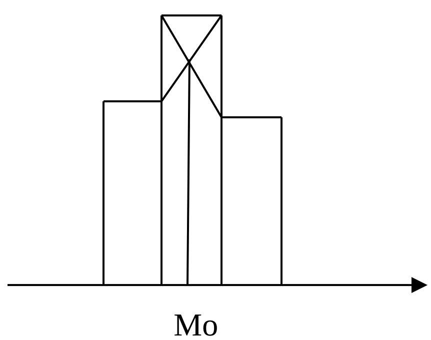
$\Delta_2 =$ The difference between the frequency of the modal class and the frequency of the post-modal class.

i= The size of the modal class.

Example 12:
The following data relate to the sales of 100 companies:

Sales(Tk.lakhs)	**No. of companies**
Below 60	12
60-62	18
62-64	25
64-66	30
66-68	10
68-70	3
70-72	2

Calculate the **modal sales**.



Example 13:
The daily profits in TK's of 100 shops are given as follows:

Profits(in Tk's)	**No. of shop**
0-100	12
100-200	18
200-300	27
300-400	20
400-500	17
500-600	6

Draw the histogram and then find the modal value. Check this value by direct calculation.

Problem 14: For Two Numbers a and b Prove that, $AM \geq GM \geq HM$

Proof: Suppose that a and b are two observations.

$$AM = \frac{a+b}{2}$$

$$GM = (ab)^{\frac{1}{2}} = \sqrt{ab} = \sqrt{a}\sqrt{b}$$

$$HM = \frac{1}{\frac{1}{2}(\frac{1}{a}+\frac{1}{b})} = \frac{2ab}{a+b}$$

$$\text{Now } AM - GM = \frac{a+b}{2} - \sqrt{a}\sqrt{b}$$

$$= \frac{1}{2}\left(a + b - 2\sqrt{a}\sqrt{b}\right)$$

$$= \frac{1}{2}\left(\left(\sqrt{a}\right)^2 + \left(\sqrt{b}\right)^2 - 2\sqrt{a}\sqrt{b}\right)$$

$$= \frac{1}{2}(\sqrt{a} - \sqrt{b})^2 \geq 0$$

[Square of any number is positive]

Thus, $AM - GM \geq 0$

$$AM \geq GM$$

Again $GM - HM$

$$= \sqrt{a}\sqrt{b} - \frac{2ab}{a+b}$$

$$= \frac{1}{a+b}\left(\sqrt{a}\sqrt{b}(a+b) - 2ab\right)$$

$$= \frac{\sqrt{a}\sqrt{b}}{a+b}\left(a + b - 2\sqrt{a}\sqrt{b}\right)$$

$$= \frac{\sqrt{a}\sqrt{b}}{a+b}\left(\left(\sqrt{a}\right)^2 + \left(\sqrt{b}\right)^2 - 2\sqrt{a}\sqrt{b}\right)$$

$$= \frac{\sqrt{a}\sqrt{b}}{a+b}(\sqrt{a} - \sqrt{b})^2 \geq 0$$

Since, $a > 0,\ b > 0, \left(\sqrt{a} - \sqrt{b}\right)^2 > 0$, so

$$GM - HM \geq 0$$

$$\Rightarrow GM \geq HM$$

$$\therefore AM \geq GM \geq HM$$

Problem 1: The following table shows the distribution of marks obtained by students in a test:

Marks	Number of Students
0 – 10	5
10 – 20	8
20 – 30	12
30 – 40	7
40 – 50	3

Calculate the mean marks of the students.

Problem 2: A survey was conducted to find the number of hours teenagers spend on social media per week. The data is presented below:

Hours	Frequency
0 – 5	4
5 – 10	6
10 – 15	10
15 – 20	12
20 – 25	8

Determine the mean number of hours spent on social media.

Problem 3: The heights of students in a school are grouped as follows:

Height (cm)	Frequency
120 – 130	10
130 – 140	14
140 – 150	16
150 – 160	8
160 – 170	2

Calculate the median height of the students.

Problem 4: The following table shows the distribution of monthly incomes of families in a town:

Income (in $)	Frequency
2000 – 3000	20
3000 – 4000	25
4000 – 5000	15
5000 – 6000	10
6000 – 7000	5

Determine the median monthly income.

Problem 5: The following table shows the scores of students in a mathematics exam:

Scores	Frequency
0 – 20	3
20 – 40	7
40 – 60	12
60 – 80	18
80 – 100	10

Calculate the mode of the scores.

Problem 6: A shopkeeper recorded the sales of different quantities of an item in a week:

Quantity Sold	Frequency
0 – 10	5
10 – 20	9
20 – 30	15
30 – 40	10
40 – 50	6

Determine the mode of the quantity sold.

Example 2: Find the mean, mode, and median for the following data,

Class	0-10	10-20	20-30	30-40	40-50	Total
Frequency	8	16	36	34	6	100

Measures of Dispersion

❖ **Dispersion**:

The literal meaning of dispersion is **scatteredness**. The measurement of the scatter of the values of a data set among themselves is called a measure of dispersion or variation. It tells us how compactly the individual values are distributed around themselves.

The study of dispersion bears its importance from the fact that various distributions may have the same averages, but substantial differences in variability.

Example:

Let the scores of two batches each of size 4, be as follows:

Batch I	49	50	50	51
Batch II	0	0	100	100

❖ **Different measures of Dispersion**:

There are several methods of measuring dispersion. These measures can be divided into two groups:

- Absolute measure
- Relative measure

❖ **Absolute measure**:

The measures are absolute in the sense that they are expressed in the same statistical units in which the original data is presented. These values may be used to compare the variation in two or more than two distributions provided the variables are expressed in the same units and have almost the same average value.

Following are the absolute measures of variation or dispersion

- Range
- Quartile Deviation
- Mean Deviation
- **Standard Deviation**

❖ **Relative Measures of Variation**:

A measure of relative variation is the ratio of a measure of absolute variation to an average. Relative measures may also be used to compare the relative accuracy of data.
The following are the relative measures of variation:

- Co-efficient of range
- Co-efficient of quartile deviation
- Co-efficient of mean deviation
- **Co-efficient of standard deviation**
- **Co-efficient of variation**

- **Range:**

Range is the simplest method of studying variation. It is defined as the difference between the value of the smallest observation and the value of the largest observation included in the distribution. Symbolically,

$$R = L - S$$

Where, $L =$ Largest value and $S =$ Smallest value

The relative measures corresponding to range, called the **co-efficient of range**, is obtained by applying the following formula

$$\textbf{Co-efficient of Range} = \frac{L - S}{L + S}$$

In a frequency distribution, the range is calculated by taking the difference between the lower limit of the lower class and the upper limit of the highest class.

Example # 1:
The following are the points of shares of a company from Monday to Saturday:

Day	Prices (Tk.)
Monday	200
Tuesday	210
Wednesday	208
Thursday	160
Friday	220
Saturday	250

Calculate range and co-efficient of range.

Solution:
We know that, Range $R = L - S$

Here, $L = 250$ and $S = 160$

$\therefore$ Range = 250-160 = Tk. 90

Co-efficient of Range $= \frac{L-S}{L+S} = \frac{250-160}{250+160}$ = 0.219

Example # 2:
Calculate the coefficient of range and range from the following data:

Profits (Tk. lakhs)	No. of Companies.
10-20	8
20-30	10
30-40	12
40-50	8
50-60	4

Solution:
In a frequency distribution, range is calculated by taking the difference between the lower limit of the lower class and the upper limit of the highest class.

Range $= L - S$ = 60-10 =50

Co-efficient of Range $= \frac{L-S}{L+S} = \frac{60-10}{60+10} = \frac{50}{70}$ = 0.714

Example # 3:
The following are the wages of 8 workers of a factory. Find the range of variation and also compute the coefficient of range. Wages in Tk.

1400 1450 1520 1380 1485 1495 1575 1440

Solution:

Range $= L - S$

Where, $L =$ Largest value and $S =$ Smallest value

Hence, $L = 1575$ and $S = 1380$

$\therefore$ Range $= 1575 - 1380$ =Tk.195

Example # 4:
The following are the marks of 80 students in a class. Find the range of variation of marks and also compute the co-efficient of range.

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of Students	4	12	20	18	15	8	2	1

Solution:

Range $= L - S$
Here, L=80, S=0
$\therefore$ Range = 80-0
= 80

Co-efficient of Range $= \frac{L-S}{L+S} = \frac{80-0}{80+0} = 1$

❖ **Mean deviation:**

Mean deviation is obtained by calculating the absolute deviations of each observation from mean (or median or mode) and then averaging these deviations by taking their arithmetic mean.

Let $X_1, X_2, ..., X_n$ be n observations of a variable with mean ($\bar{x}$), median (M_e) and Mode (M_o) then mean deviation is defined by:

<u>For Ungrouped Data</u>:

$$M.D_{(\bar{x})} = \frac{1}{n}\sum\left|x-\bar{x}\right| \qquad \text{[concerning arithmetic mean]}$$

$$M.D_{(M_e)} = \frac{1}{n}\sum\left|x-M_e\right| \qquad \text{[using medium]}$$

$$M.D_{(M_o)} = \frac{1}{n}\sum\left|x-M_o\right| \qquad \text{[using mode]}$$

<u>For grouped Data</u>:

$$M.D_{(\bar{x})} = \frac{1}{n}\sum f\left|x-\bar{x}\right|$$

$$M.D_{(M_e)} = \frac{1}{n}\sum f\left|x-M_e\right|$$

$$M.D_{(M_o)} = \frac{1}{n}\sum f\left|x-M_o\right|$$

Where, $n = \sum f$

Co-efficient of mean deviation is the ratio of the mean deviation measured from a certain measure of central location to the corresponding measure of central location and is defined as follows:

$$C.M.D_{(\bar{x})} = \frac{M.D_{\bar{x}}}{\bar{x}}$$

$$C.M.D_{(M_e)} = \frac{M.D_{(M_e)}}{M_e}$$

$$C.M.D_{(M_o)} = \frac{M.D_{(M_o)}}{M_o}$$

<u>Problem-5</u>:

Calculate the **mean deviation** from

- **(a)** Arithmetic mean
- **(b)** Mode
- **(c)** Median

In respect of the marks obtained by nine students given below show that the mean deviation from the median is minimum.

Marks (Out of 25)	7	4	10	9	15	12	7	9	7

<u>Solution</u>:

We know that,

$$M.D_{(\bar{x})} = \frac{1}{n}\sum|x-\bar{x}|$$

$$M.D_{(M_e)} = \frac{1}{n}\sum|x-M_e|$$

$$M.D_{(M_o)} = \frac{1}{n}\sum|x-M_o|$$

Now,

Mean, $\bar{x} = \frac{\sum x_i}{n} = \frac{80}{9} = 8.89$

For calculating the **median** the items have to be arranged

Marks (Out of 25)	4	7	7	7	9	10	10	12	15

$\therefore$ Median = size of $\frac{n+1}{2}$ th item = $\frac{9+1}{2}$ = 5th item

Here, the size of 5^{th} item = 9.

Hence, median=9.

Mode = 7 (since 7 is repeated as the maximum number of items i.e. 3)

Calculation of Mean Deviation:

Marks x	**Deviations from Mean =** $\lvert x_i - \bar{x} \rvert$	**Deviations from Median =** $\lvert x_i - Me \rvert$	**Deviations from Mode =** $\lvert x_i - Mo \rvert$
7	1.89	2	0
4	4.89	5	3
10	1.11	1	3
9	0.11	0	2
15	6.11	6	8
12	3.11	3	5
7	1.89	2	0
9	0.11	0	2
7	1.89	2	0
Total	$\sum \lvert x_i - \bar{x} \rvert = 21.11$	$\sum \lvert x_i - M_e \rvert = 21$	$\sum \lvert x_i - M_o \rvert = 23$

Now we have,

$$M.D_{(\bar{x})} = \frac{1}{n}\sum \lvert x - \bar{x} \rvert = \frac{1}{9} \times 21.11 = 2.34$$

$$M.D_{(M_e)} = \frac{1}{n}\sum \lvert x - M_e \rvert = \frac{1}{9} \times 21 = 2.33$$

$$M.D_{(M_o)} = \frac{1}{n}\sum \lvert x - M_o \rvert = \frac{1}{9} \times 23 = 2.56$$

From these calculations, it is clear that the mean deviation is the least from the median.

<u>Problem-6</u>:
Calculate mean deviation (taking deviations from mean) from the following data:

x	2	4	6	8	10
f	1	4	6	4	1

<u>Solution</u>: We know that,

$$M.D_{(\bar{x})} = \frac{1}{n}\sum f\left|x-\bar{x}\right|$$

Now,

$$\bar{x} = \frac{\sum fx}{n}$$

By using a calculator we get, $\sum fx$ = 96 and $n = \sum f$ = 16.

$$\therefore \bar{x} = \frac{\sum fx}{n} = \frac{96}{16} = 6$$

Calculation of mean deviation

x	f	**Deviations from Mean =** $\left\lvert x-\bar{x}\right\rvert$	$f\left\lvert x-\bar{x}\right\rvert$
2	1	4	4
4	4	2	8
6	6	0	0
8	4	2	8
10	1	4	4
Total			$\sum f\left\lvert x-\bar{x}\right\rvert = 24$

$$\therefore\ M.D_{(\bar{x})} = \frac{1}{n}\sum f\left|x-\bar{x}\right|$$

$$= \frac{1}{16}\times 24$$

$$= 1.5$$

<u>**Problem-7**</u>**:**

Calculate mean deviation (**from mean**) from the following data:

Size of the item	3-4	4-5	5-6	6-7	7-8	8-9	9-10
Frequency	3	7	22	60	85	32	8

Solution: We know that,

$$M.D_{(\bar{x})} = \frac{1}{n}\sum f\left|x-\bar{x}\right|$$

Calculation of mean deviation from mean

Size of the item	**Frequency** f	**Mid point** x	**Deviations from Mean =** $\left\|x-\bar{x}\right\|$	$f\left\|x-\bar{x}\right\|$
3-4	3	3.5	3.59	10.77
4-5	7	4.5	2.59	18.13
5-6	22	5.5	1.59	34.98
6-7	60	6.5	0.59	35.4
7-8	85	7.5	0.41	34.85
8-9	32	8.5	1.41	45.12
9-10	8	9.5	2.41	19.28
Total	$n=\sum f = 217$			$\sum f\left\|x_i-\bar{x}\right\|$ **=198.53**

Now we have,

$$\bar{x} = \frac{\sum fx}{n} = \frac{1538.5}{217} = 7.09$$

$$\therefore\ M.D_{(\bar{x})} = \frac{1}{n}\sum f\left|x-\bar{x}\right|$$

$$= \frac{1}{217}\times 198.53 = 0.9148 = 0.915$$

Problem-8:
Age distribution of hundred life insurance policy holders is as follows:

Age as on nearest birthday	Number
17-20	9
20-26	16
26-36	12
36-41	26
41-51	14
51-56	12
56-61	6
61-71	5

Calculate **mean deviation from median** age.

Solution: We know that,

$$M.D_{(M_e)} = \frac{1}{n}\sum f\,|x - M_e|$$

Calculation of mean deviation

Age as on nearest birthday	**Number** f	$c.f$	**Midpoint** x	**Deviations from Median =** $\lvert x-M_e \rvert$	$f\lvert x-M_e \rvert$
17-20	9	9	18.5	19.75	177.75
20-26	16	25	23	15.25	244.00
26-36	12	37	31	7.25	87.00
36-41	26	63	38.5	0.25	6.50
41-51	14	77	46	7.75	108.50
51-56	12	89	53.5	15.25	183.00
56-61	6	95	58.5	20.25	121.50
61-71	5	100	66	27.75	138.75
Total	$\sum f_i$ =100				$\sum f\lvert x-M_e \rvert$ =1067

We know that,

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} \times i$$

Here, $\frac{N}{2} = \frac{100}{2}$ = 50

Here median class is 36-41.

$$\therefore L = 36, p.c.f = 37, f = 26, i = 5$$

So we have,

$$\text{Median} = 36 + \frac{50-37}{26} \times 5 = 38.25$$

$$\therefore M.D_{(M_e)} = \frac{1}{n}\sum f\left|x - M_e\right| = \frac{1}{100} \times 1067 = 10.67$$

- ❖ **<u>Standard deviation</u>:**

Standard deviation may be defined as the positive square root of the arithmetic mean of the squares of deviations of given observations from their arithmetic mean.

<u>For ungrouped data</u>:

Let, $x_1, x_2, ..., x_n$ denote n values of a variable x. The standard deviation denoted by S_x is defined as

$$S_x = \sqrt{\frac{\sum(x-\bar{x})^2}{n}} = \sqrt{\left\{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2\right\}} = \sqrt{\left\{\frac{\sum x^2}{n} - \left(\bar{x}\right)^2\right\}}$$

Where, $\bar{x} = \frac{\sum x}{n}$.

<u>For grouped data</u>:

If $x_1, x_2, ..., x_n$ occurs with frequency $f_1, f_2, ..., f_n$ respectively, the standard deviation is defined as

$$S_x = \sqrt{\frac{\sum f\left(x-\bar{x}\right)^2}{N}} = \sqrt{\left\{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2\right\}} = \sqrt{\left\{\frac{\sum fx^2}{N} - \left(\bar{x}\right)^2\right\}}$$

Where $N = \sum f$ and $\bar{x}$ is the arithmetic mean of the distribution.

Problem-9:

Find the **standard deviation** from the weekly wages of ten workers working in a factory:

Workers	wages	workers	wages
A	320	F	340
B	310	G	325
C	315	H	321
D	322	I	320
E	326	J	331

Solution:

We know that,

$$S_x = \sqrt{\left\{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2\right\}}$$

By using a calculator we get,

$$\sum x^2 = 1043912,\ \sum x = 3230 \text{ and } n = 10.$$

So that,

$$S_x = \sqrt{\left\{\frac{1043912}{10} - \left(\frac{3230}{10}\right)^2\right\}} = 7.886 = 7.89$$

Example: There are 39 plants in the garden. A few plants were selected randomly and their heights in cm were recorded as follows: 51, 38, 79, 46, 57. Calculate the variance and standard deviation of their heights.

Problem-10:

An analysis of production rejects resulted in the following figures:

No. of rejects per operator	No. of operators
21-25	5
26-30	15
31-35	28
36-40	42
41-45	15

46-50	12
51-55	3

Solution:
We know that

$$S_x = \sqrt{\left\{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2\right\}}$$

Where, $N = \sum f$.

Calculation of Standard Deviation

No. of rejects per operator	**No. of operators** f	**Midpoint** x	x^2	fx	fx^2
21-25	5	23	529	115	2645
26-30	15	28	784	420	11760
31-35	28	33	1089	924	30492
36-40	42	38	1444	1596	60648
41-45	15	43	1849	645	27735
46-50	12	48	2304	576	27648
51-55	3	53	2809	159	8427
Total	120		10808	4435	169355

From the data by using a calculator we get,

$\sum fx = 4435$, $N = \sum f$ =120 and $\sum f x^2 = 169355$

$$\therefore S_x = \sqrt{\left\{\frac{169355}{120} - \left(\frac{4435}{120}\right)^2\right\}}$$

= 6.7359
= 6.74

❖ **Coefficient of Standard Deviation:**

Co-efficient of standard deviation is defined by

$$C.S.D = \frac{S.D}{Mean} = \frac{\sigma}{\bar{x}} = \frac{s}{\bar{x}}$$

Coefficient of Variation:

The relative measure of dispersion based upon standard deviation is called co-efficient of standard deviation. The coefficient of standard deviation multiplied by 100 gives the coefficient of variation.

Thus, Co-efficient of variation (C.V)

$$= \frac{\text{Standard deviation}}{\text{Mean}} \times 100$$

$$= \frac{\sigma}{\bar{x}} \times 100$$

where, σ and $\bar{x}$ are both measured in the same units.

Example-11:

Calculate co-efficient of variation from the following data:

Profits (Rs. crores)	10-20	20-30	30-40	40-50	50-60
No. of Companies	8	12	20	6	4

Solution:

We know that,

$$C.V = \frac{\sigma}{\bar{x}} \times 100 \qquad \text{and } \bar{x} = \frac{\sum fx}{N}$$

$$\sigma = \sqrt{\left\{ \frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N} \right)^2 \right\}}$$

Calculation of coefficient of variation

Profits (Tk. crores)	**Midpoint** (x)	**No. of cost** (f)	x^2	fx	fx^2
10-20	15	8	225	120	1800
20-30	25	12	625	300	7500
30-40	35	20	1225	700	24500
40-50	45	6	2025	270	12150
50-60	55	4	3025	220	12100
Total		50	7125	1610	58050

By using a calculator we get

$\sum fx = 1610$, $\sum fx^2 = 58050$ and $N = \sum f = 50$.

$$\therefore \bar{x} = \frac{\sum fx}{N} = \frac{1610}{50} = 32.2$$

$$\therefore \sigma = \sqrt{\left\{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2\right\}} = \sqrt{\left\{\frac{58050}{50} - \left(\frac{1610}{50}\right)^2\right\}} = 11.14$$

$\therefore$ Co-efficient of variation $= \frac{\sigma}{\bar{x}} \times 100 = \frac{11.14}{32.2} \times 100 = 34.596 = 34.6\%$

<u>Example-12</u>:

A purchasing agent obtained samples of 60-watt bulbs from two companies. He had the samples tested in his laboratory for length of life with the following result:

Length of life (in hours)	Company A	Company B
1700-1900	10	3
1900-2100	16	40
2100-2300	20	12
2300-2500	8	3
2500-2700	6	2

a) Which company's bulbs do you think are better?

b) If prices of both types are the same, which company's bulbs would you buy?

Solution:

Calculation of Mean and coefficient of variation

Length of life (in hours)	Midpoint	Company A	Company B
1700-1900	1800	10	3
1900-2100	2000	16	40
2100-2300	2200	20	12
2300-2500	2400	8	3
2500-2700	2600	6	2

For Company A:

By using a calculator we get

$$\sum fx = 128800,\ \sum fx^2 = 279840000 \text{ and } N = \sum f = 60$$

$$\therefore\ \bar{x} = \frac{\sum fx}{N} = \frac{128800}{60} = 2146.67 \text{ and}$$

$$\sigma = \sqrt{\left\{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2\right\}} = \sqrt{\left\{\frac{279840000}{60} - \left(\frac{128800}{60}\right)^2\right\}} = 236.267 = 236.27$$

$$\therefore \text{Co-efficient of variation} = \frac{\sigma}{\bar{x}} \times 100 = \frac{236.27}{2146.67} \times 100 = 11.00 = 11\%$$

For Company B:

By using a calculator we get

$$\sum fx = 124200,\ \sum fx^2 = 258600000 \text{ and } N = \sum f = 60$$

$$\therefore\ \bar{x} = \frac{\sum fx}{N} = \frac{124200}{60} = 2070 \text{ and}$$

$$\sigma = \sqrt{\left\{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2\right\}} = \sqrt{\left\{\frac{258600000}{60} - \left(\frac{124200}{60}\right)^2\right\}} = 158.429 = 158.43$$

$$\therefore \text{Co-efficient of variation} = \frac{\sigma}{\bar{x}} \times 100 = \frac{158.43}{2070} \times 100 = 7.65\%$$

(a) Since the average is higher in the case of company A, hence bulbs of company A are better.

(b) The Coefficient of variation is lower for company B. Hence if prices are the same we will prefer to buy Company B's bulbs.

<u>Example-13</u>:
The prices of a Tea company shares in Dhaka and Chittagong Markets during the last ten months are recorded below:

Month	**Dhaka**	**Chittagong**
January	105	108
February	120	117
March	115	120
April	118	130
May	130	100
June	127	125
July	109	125
August	110	120
September	104	110
October	112	135

Determine the arithmetic mean and standard deviation of the prices of shares. In which market are the share prices stable?

<u>Solution</u>:
For determining in which market prices of shares are more stable we shall compare the coefficient of variation. Let prices in Dhaka and Chittagong are denoted by X and Y respectively:

<u>Dhaka</u>:

$$\overline{X} = \frac{\sum X}{N} = \frac{1150}{10} = 115$$

$$\sigma = \sqrt{\left\{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2\right\}} = 8.33$$

$$C.V = \frac{\sigma}{\overline{X}} \times 100 = \frac{8.33}{115} \times 100 = 7.24\%$$

Chittagong:

$$\overline{X} = \frac{\sum X}{N} = \frac{1190}{10} = 119$$

$$\sigma = \sqrt{\left\{\frac{\sum X^2}{N} - \left(\frac{\sum X}{N}\right)^2\right\}} = 10.089 = 10.09$$

$$C.V = \frac{\sigma}{\overline{X}} \times 100 = \frac{10.09}{119} \times 100 = 8.478 = 8.48\%$$

Since the coefficient of variation is less in Dhaka, hence the share price in the Dhaka market shows greater stability.

❖ **Find the CV of the following series: 152, 154, 156, ………,200.**

Ans: Let $U = \frac{x-150}{2}$ = 1, 2, 3, ………, 25

We know, for first n natural numbers,

$$\sigma^2 = \frac{n^2-1}{12}$$

$$\therefore \sigma_u{}^2 = \frac{25^2-1}{12} = 52$$

Also we know that σ^2 depends on scale but independent of origin.

So, $\sigma_x{}^2 = \sigma_u{}^2 \times c^2$ here, c= 2

$= 52 \times 2^2$

$$\Rightarrow \sigma_x = \sqrt{4 \times 13 \times 4} = 4\sqrt{13}$$

$\therefore CV = \frac{\sigma_x}{\bar{x}} \times 100$ where, $\bar{x} = \frac{20(20+1)}{2} = 210$

$= \frac{4\sqrt{13}}{210} \times 100 = 6.87\%$

❖ **Find the CV of the following series: 510, 520, 530, 540, 550, 560, 570.**

<u>Ans</u>: Let $U = \frac{x-500}{10}$ = 1, 2, 3, 4, 5, 6, 7

We know, for first n natural numbers,

$$\sigma^2 = \frac{n^2-1}{12}$$

$$\therefore \sigma_u{}^2 = \frac{7^2-1}{12} = 4$$

Also we know that σ^2 depends on scale but independent of origin.

So, $\sigma_x{}^2 = \sigma_u{}^2 \times c^2$ here, c= 10

$= 4 \times 10^2 = 400$

$\Rightarrow \sigma_x = \sqrt{400} = 20$

$\therefore CV = \frac{\sigma_x}{\bar{x}} \times 100$ where, $\bar{x} = \frac{7(7+1)}{2} = 28$

$= \frac{20}{28} \times 100$

$= 71.43\%$

Formula for calculating sample standard deviation

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Population Standard Deviation	$\sigma = \sqrt{\frac{\Sigma (x_i - \mu)^2}{n}}$	μ = Population Average x = Individual values in population n = Count of values in population
Sample Standard Deviation	$S = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{(n-1)}}$	$\bar{x}$ = Sample Average x = Individual values in sample n = Count of values in sample

Example: For the following data, determine the mean, variance, and standard deviation:

Population Standard Deviation	Sample Standard Deviation
$\sigma = \sqrt{\dfrac{\Sigma(x_i - \mu)^2}{N}}$	$s = \sqrt{\dfrac{\Sigma(x_i - \bar{x})^2}{n-1}}$
Where: σ is the population standard deviation $\sqrt{}$ is the symbol for taking the square root $\sum$ is the symbol for summation, indicating that you should take the sum of everything that follows it x_i is a particular value in your data μ is the population mean $(x_i - \mu)$ is the distance between a particular value in your population data and the mean N is the population size	Where: s is the population standard deviation $\sqrt{}$ is the symbol for taking the square root $\sum$ is the symbol for summation indicating, that you should take the sum of everything that follows it x_i is a particular value in your data $\bar{x}$ is the sample mean $(x_i - \bar{x})$ is the distance between a particular value in your sample data and the mean n is the sample size*

*When calculating the sample standard deviation, we divide by n-1 to get a closer approximation of the true population standard deviation,σ.

Class Interval	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	27	10	7	5	4	2

Solution :

Class Interval	Frequency (f)	Mid Value (xi)	fxi	fxi2
0-10	27	5	135	675
10-20	10	15	150	2250
20-30	7	25	175	4375
30-40	5	35	175	6125
40-50	4	45	180	8100
50-60	2	55	110	6050
	∑f=55		∑fxi= 925	∑fxi2= 27575

- N = ∑f = 55
- Mean = (∑fxi)/N = 925/55 = 16.818
- Variance = 1/(N – 1)[∑fxi2 – 1/N(∑fxi)2]
- = 1/(55 – 1) [27575 – (1/55) (925)2]
- = (1/54) [27575 – 15556.8182]
- = 222.559
- Standard deviation = √variance = √222.559 = 14.918